

The Effects of Physical Context on Collaborator Placement in Augmented Reality Meetings

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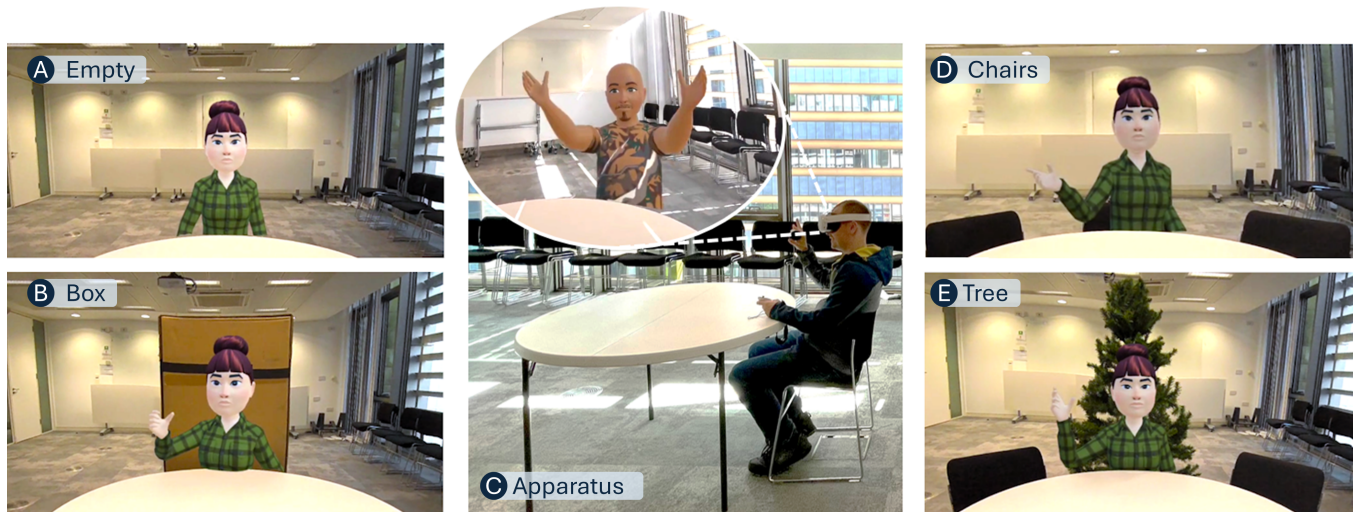


Figure 1: We conducted two user studies to explore how local users position a remote collaborator’s avatar in AR (C) based on physical context. In the first participants arranged avatars in an empty setting (A) or one featuring a large box (B). In the second study, avatars were placed in contexts featuring chairs (D) or a tree (E). Sub-figures (B) and (E) highlight physicality conflicts, where the placement of the avatar clashed with the presence of physical objects.

Abstract

Everyday Augmented Reality (AR) holds great potential for transforming collaborative practices, but many aspects remain under-explored, particularly how local users position remote collaborators within their real, often cluttered, environments. We conducted two studies to explore collaborator avatar placement preferences in AR,

examining the effects of various physical objects (chairs, box, tree). Participants preferred avatar placements that were more frontal due to their ergonomics and superior collaborator visibility, corresponding to locations where a real person would sit in a similar physical environment. Chairs invited placement, while the tree was avoided. Notably, the large and visually simple box elicited mixed reactions, being viewed alternately as an obstacle or a physical anchor for the avatars. These findings underscore the need for AR collaboration platform designs that balance realism with user-driven flexibility, ensuring that social layouts align with both physical contexts and user preferences.

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CCS Concepts

• **Human-centered computing** → **Empirical studies in collaborative and social computing**; *User studies*; **Mixed / augmented reality**.

Keywords

Augmented Reality, Mixed Reality, Collaboration, Context-awareness, Affordance, Anti-affordance, F-Formations, Physicality Conflict, Thematic Analysis

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1 Introduction and Background

Everyday Augmented Reality (AR) headsets are a transformative technology poised to revolutionise collaboration across many domains [6, 24]. Designed for use in daily life - whether at work, home, or on the go - they facilitate interaction with both local and remote participants, as well as real and virtual artifacts. Headsets such as the Apple Vision Pro¹ and the Meta Quest 3² come with collaboration platforms such as FaceTime³ and Meta Horizon Workrooms⁴.

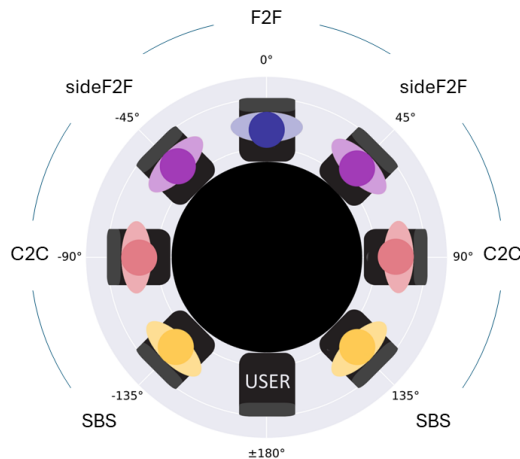


Figure 2: Kendon’s F-Formations on a round table: Face-to-Face (F2F), Corner-to-Corner (C2C), Side-by-Side (SBS). We added two further formations that respect the equal spacing of 45° between formations. These are sideF2F, equidistant from F2F and C2C.

¹<https://www.apple.com/uk/apple-vision-pro/>

²<https://www.meta.com/gb/quest/quest-3/>

³<https://support.apple.com/en-gb/guide/apple-vision-pro/tan440238696/visionos>

⁴<https://forwork.meta.com/gb/horizon-workrooms/>

In real life, two or more people can naturally group themselves into various patterns called *facing formations* (F-Formations), see Fig. 2, such as face-to-face (F2F, also known as vis-à-vis), corner-to-corner (C2C, also known as L-arrangement) and side-by-side (SBS) [29], which are often dynamic [53] and affected by several environmental factors [9, 23, 25, 26]. However, current AR meeting platforms typically position remote collaborators using preset configurations, often defaulting to face-to-face arrangements, and may not allow users to manually re-position collaborators. Extensive foundational literature emphasizing the importance of Inter-Personal Distances (IPDs) [17] and seating arrangements [29] on real-life social behaviour highlighted a range of layout preferences influenced by the environment [8, 11, 16, 49–51]. This suggests that relying solely on face-to-face positioning is insufficient. Additionally, recent studies on AR avatar placement show that users dislike manually adjusting their interlocutor’s position [47], underscoring the need for automated systems that intelligently place remote users in appropriate social layouts based on situational factors. This paper thus investigates user social layout preferences.

While collaboration in AR has been studied, and platforms have been created to attempt to match remote users’ different physical environments [28, 45, 46], a key issue remains: the optimal positioning of remote participants within the real environment of the local user. The placement of virtual content in AR is influenced by physical surroundings and informed by conflicts with real-world objects [3, 13, 28, 30, 31, 44, 52], by the semantic coherence between content and setting [10, 12, 34, 36, 37], and by social etiquette and dynamics [4, 10, 39]. While these studies suggest the physical context may affect remote collaborator placement, few investigated the preferred positioning of remote collaborators in AR environments [32, 47, 54]. While such studies shed light on the challenges of designing collaborative spaces for remote users in AR, they do not study the effect of the physical context on preferred collaborator position.

An affordance can be defined as an “*action possibility* that is readily perceivable by an actor” and an anti-affordance as “the prevention of interaction” [43], thus determining how an object may or may not be used by an agent: a chair affords the act of sitting, while a spiked chair prevents it. Similarly, certain elements in the physical context of an AR meeting - such as couches, filing cabinets, etc. - may invite local users to - or prevent them from - placing their collaborator’s avatar in specific locations. It is unclear whether physical affordances and anti-affordances impact preferred avatar placement. It is also uncertain whether users avoid physicality conflicts with the real environment while selecting their preferred social layouts. This paper investigates user preferences and criteria for dealing with physical affordances and anti-affordances while placing their collaborator’s avatar.

We conducted two within-subjects studies to explore how physical contexts influence the placement of virtual collaborators in AR environments. The first study ($N = 34$) compared placements in a meeting room with and without a large box in the local user’s space. It revealed a preference for placing avatars in frontal or slightly off-centre positions, with ergonomics and collaborator visibility shaping decisions. While the box clearly influenced user preferences, it received mixed reactions as to whether it acted as an affordance or anti-affordance. The second study ($N = 24$) built

Physical Environment Conditions

Study 1 Conditions



Study 2 Conditions

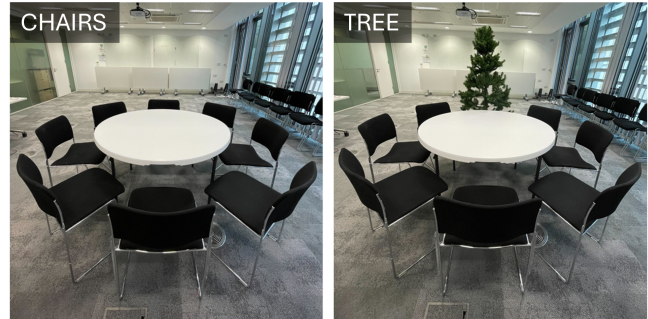


Figure 3: The physical environment conditions in the two studies. In the first, we compared an *Empty* context with one where a *Box* was placed in the participant's preferred avatar location. In the second, we compared a context providing *Chairs* around the table with one where one of the chairs was replaced by a *Tree*. The tree was placed in the participant's preferred avatar location.

on these findings by comparing the effect of the presence of different physical objects on avatar placement, finding that chairs invited avatar positioning, while a fake plant was perceived as an anti-affordance and avoided.

2 User Study 1 - Understanding Avatar Placement Preferences

We conducted a user study to identify where users prefer their interlocutor's avatar to be positioned and understand the rationale behind these preferences, and to examine whether the physical context influences these placement decisions and why.

2.1 AR Application

The study utilized two Meta Quest 3 headsets running a collaborative app developed in Unity. Remote users were embodied with customizable Meta avatars⁵ featuring body and lip tracking. Connectivity was established using Photon Fusion⁶ over WiFi. This setup replicated XR platforms like Meta Horizon Workrooms⁷ and Microsoft Mesh⁸.

2.2 Methodology

Pairs of participants collaborated from twin rooms designed to emulate professional work environments. The rooms were spacious and quiet, as shown in Figure 3. Each user sat at a round table with a 1.2m diameter allowing comfortable social distances with collaborators [7, 8, 18, 20–22, 35, 42] and allowing all f-formations described by Kendon [29], including an additional two *sideF2F* formations which offset F2F by $\pm 45^\circ$ to the left and right, shown in Figure 2. These positions were included based on Reinhardt's findings, which highlighted a preference among some participants for slightly off-centre arrangements [47]. Participants completed a mixed-motive

negotiation task [38], which involved debating who should receive a work-related benefit, such as "*a new computer*" or "*a £10,000 salary bonus*". Participants received a £10 Amazon voucher for their participation, with an additional £5 vouchers awarded to participants who convinced their collaborator they should receive the most valuable benefits, thus fostering competition [19]. This type of task, used in related earlier studies to encourage collaboration [1, 26, 27, 32], involves participants cooperating to resolve a conflict of interest and reach an agreement [38].

We recruited 36 participants (mean age = 27.4, SD = 7.0, male = 21, female = 14, non-binary = 1) of various familiarity with AR interfaces (42% never using it) and high familiarity with video-conferencing platforms (89% using it at least once a week). Only pairs of acquaintances or friends were accepted, as prior research shows interpersonal intimacy affects interaction distance [33], with strangers favouring greater distances [8, 35] and romantic partners preferring intimate spaces [17].

After signing consent forms and completing demographic surveys, each participant was asked to go to their room and use the custom app to "*place your collaborator's avatar in whatever position around the table you prefer the most*". The study involved two physical environment conditions where the space around the table was first *Empty* (no additional objects) and later included a *Box*. This large, brown, featureless box was placed in the location selected by each user as their preferred spot around the table from the *Empty* condition. The box was $150 \times 70 \times 20\text{cm}$, occupying a space larger than the virtual avatar, and was chosen as its presence could be generalised to other common office items, such as dividers, filing cabinets, or other physical barriers. Users adjusted avatar placement multiple times during the 3 minute negotiation task. Avatar placements reset every 36 seconds to encourage reconsideration, and control alternated between participants, ensuring a minimum of 5 placements per user per condition.

We conducted a semi-structured interview after each session to investigate each participant's preferences and the rationales

⁵<https://www.meta.com/gb/avatars/>

⁶<https://doc.photonengine.com/fusion/v1/fusion-intro>

⁷<https://forwork.meta.com/gb/horizon-workrooms/>

⁸<https://www.microsoft.com/en-us/microsoft-teams/microsoft-mesh>

behind them. Topics tackled included placement preference (*When there wasn't a box in the environment, where did you place your collaborator?*) and influence on said preference by the physical props (*Did the presence of the box impact your preference?*).

2.3 Results

We conducted an inductive, data-driven thematic analysis of the post-study interviews, focusing on user rationale behind collaborator placements. We employed an open coding approach, followed by axial coding, to identify and categorise relevant themes and patterns within the interview recordings [48]. These were auto-transcribed using WhisperX⁹, then reviewed, corrected when necessary, and anonymised. Two researchers independently familiarised themselves with the data and created individual coding schemes using NVivo¹⁰, analysing the data line by line. The researchers then compared, refined, and merged their coding schemes. After developing higher-level codes by grouping related codes, the researchers iteratively discussed and integrated these codes into the following themes.

Preferred F-Formations. Most ($N = 34$) participants preferred the layouts that positioned their collaborator across the table: F2F and sideF2F. Of these, many ($N = 23$) reported finding them the most comfortable: *"It was just optimising for laziness and physical exertion. It's easier to look straight ahead"* (P6). P23 hypothesized that: *"The weight of the headset makes that neck turn more difficult, and also more dramatic because your visual field is reduced. So, whilst in real life I need to turn just a bit to look at you, if we were wearing the [headset] I would have to completely turn my head to have you in my visual field - being sat so close to me"*. Collaborator visibility was thus another important factor, mentioned by 20 participants: *"I always placed them in front because it's the easiest way to communicate: you can see the body language, the posture"* (P9). On the other hand, 14 participants mentioned that the frontal position, F2F, was *"a bit intimidating"* (P27) either because it felt too formal (*"I find [it] very formal - unless you're eating dinner"*, P13), too adversarial (*"It's more confrontational because of the physical barrier: the desk"*, P30) or because *"You can go into the uncanny valley route if they're placed directly in front of you. They're not a real person, there is no proper eye contact. It's a little bit creepy"* (P19). Some participants ($N = 8$) mentioned picking C2C layouts as they offered a good middle ground between the benefits and drawbacks of F2F: *"It felt less formal but I could also see them all the time"* (P27). Four participants, however, mentioned a lack of visibility as the main drawback of these and the SBS formations: *"Off to the side of the table it's harder to notice what's going on"* (P11). Many participants ($N = 14$) mentioned feeling uncomfortable when they were sitting in one of the closer layouts (SBS or C2C) with their collaborator: *"The seating where [the avatar] was right next to me is a little bit more invasive"* (P9).

The physical object influenced the placement preference. Most participants ($N = 31$) felt that the box (see Figure 1B) influenced their placement of their collaborator. However, this influence was not straightforward since, while many users ($N = 21$) were dissuaded

from placing their collaborator in the same place as the box, perceiving it as an anti-affordance *"I tried putting [the avatar] in the box but it looked weird, so I just moved [it]"* (P7), others ($N = 10$) were instead inclined to pick that location and considered it an affordance: *"I placed them on the box so it looks like they were sitting on a chair"* (P36). There were multiple reasons why participants did not want their collaborator to be in the same place as the box. Most participants ($N = 18$) attributed it to cognitive dissonance and that *"It's a bit stupid to have two things co-located. It physically doesn't make sense"* (P32). Of these, 8 linked this dissonance to a reduced immersion: *"A person couldn't be there in real life, so it draws to attention the fact that it's extremely not real"* (P11). Some ($N = 11$) mentioned instead the visual dissonance caused by the depth mismatch as explained by P8: *"It was like an optical illusion, because [the avatar] was actually further away, and I did not like this"*. Finally, 4 participants mentioned social etiquette: *"I just thought of her as a human being - she probably doesn't want to be on this box"* (P24). Three participants mentioned other objects that they would not place the collaborator on: *"Even though it's augmented reality, because you're so immersed in it, [the placement] needs to match reality. If there was a cactus, I wouldn't have sat them on it because in real life I wouldn't sit them on a cactus"* (P39). On the other hand, 10 participants felt the box was the perfect location to place their collaborator as it felt like a physical placeholder, often mentioning it acted as seating or framing: *"I treated the cardboard as a chair because it was nicer that she had a background. It wasn't just floating in the room"* (P10). Some participants ($N = 4$) mentioned that such placement felt right due to the box having a larger but otherwise similar size and shape as the avatar: *"Because it was kind of like a person-shaped cardboard it felt like I could line them up"* (P5). Others ($N = 4$) indicated that the flat and simple surface of the box improved their experience: *"I was only really looking at [the avatar] because the box was just plain, so it was like looking at somebody against a white wall"* (P31).

3 User Study 2 - Exploring Other Physical Contexts

Two weeks after Study 1, we conducted a follow-up within-subjects user study with 24 participants (12 returning pairs, mean age = 28.0, SD = 7.1, male = 14, female = 10) to further examine collaborator placement preferences. Participants familiarity levels with AR interfaces and videoconferencing platforms were consistent with Study 1. Participants performed the same negotiation tasks as in Study 1 under two physical context conditions, shown in Figure 3. In the first, seven chairs were arranged around the table at 45° intervals to explore whether participants preferred placing avatars on chairs, reflecting interpretations from Study 1 where the box was seen as an affordance. In the second condition, a faux Christmas tree ($h = 160\text{cm} \times \varnothing = 70\text{cm}$) replaced one of the chairs to test how participants responded to placing avatars over irregular, uncomfortable objects mentioned in the interviews as possible anti-affordances, and contrasting the flat surface of the box in Study 1. The study maintained the same apparatus and task setup as Study 1. Collaborator placement preferences were recorded after the Chairs condition and used to determine the tree's position in the Tree condition. We conducted a similar post-study interview to Study

⁹<https://github.com/m-bain/whisperX>

¹⁰<https://lumivero.com/products/nvivo/>

1, but this time focused on the impact of the props (e.g. *"Did the presence of the [chairs/tree] impact your placement?"*) and also on the physical background of the study (*"Was there any other aspect of the physical surroundings that influenced your placement preference?"*). Follow-up questions enabled us to explore the thinking process behind user choices.

3.1 Results

Compared to the first study, opinions of and reactions towards the physical objects (see Figure 3) were less mixed. Most participants ($N = 20$) considered the chairs as an affordance: *"Without even thinking about it. [The avatar is] going on the chair"* (P31). They were never considered an anti-affordance. Many ($N = 18$) stated this placement either increased the realism of the interaction (*"It felt more real if they were actually sat in a chair, because that's what would really happen"*, P27) or highlighted that, if the avatar's position disregarded the chairs, it caused a conflict: *"I always position them on a chair, not in between, because that's what you're used to seeing. You see people on chairs, it breaks the illusion a bit otherwise"* (P1).

The tree (see Figure 1E) was considered an anti-affordance by 20 participants that felt they *"Couldn't fit [the avatar] there, because the tree is there"* (P1). Most participants ($N = 16$) mentioned the semantic dissonance of placing a person on a tree as their main driving factor in placement preference: *"They're not a bird. They shouldn't be sitting in a tree. That's not very realistic"* (P32). The feeling was well encapsulated by P1: *"You're so used to this inherent sense that two things cannot occupy the same physical space. And so when that rule was broken, you're totally taken out of the immersion of it"*. Fewer participants ($N = 8$) mentioned a depth mismatch as *"Clipping through the tree looks weird - it messes with perspective"* (P12). Finally, 4 users mentioned social etiquette. P28 did not place their collaborator in place of the tree, stating: *"It seems like I'm like shoving them into a tree and they're going to have a branch in their face, even though I know they can't. I was safeguarding them"*.

Most participants ($N = 18$) considered the tree more of an anti-affordance than the box. The reasons mentioned ranged from positive aspects (*"It made the environment happier, like an ornament, I didn't want to cover it"*, P9), to negative (*"It was more of a hindrance to put [the avatar] there"*, P29). From semantic reasons (*"A box is a box: it can mean different things. A tree is a tree: you cannot really take a tree as a chair"*, P30), to visual (*"Compared to the box being so flat, the tree is very 3D. Placing the low-tech avatar on it, there probably would have been glitches and I would have seen branches poking through [their] face and that could have been a distraction"*, P10), to social ones (*"The box is a blank slate while the tree is like a person. I wouldn't put [the avatar] in front of a person! Who cares about a box, but the tree is alive!"*, P33).

Several participants ($N = 17$) mentioned how the physical context of the experiment influenced their placement preference. Some ($N = 6$) mentioned the shape and size of the room: *"If you're in a big room and then not using whatever space for your conversation, it makes me feel a bit weird. Like, why are you hiding away?"* (P34). Others ($N = 11$) wanted to either face or avoid facing the window: *"I was going window side, probably not consciously, probably because it's a nicer backdrop"* (P6); *"I wouldn't put [the avatar] against the*

light because it would also hit my eyes as well" (P8). Four participants mentioned that the presence of physical objects on the table, especially if part of a collaborative task, would have affected their preferred social formation: *"If there's something on the table, I'd definitely move [the avatar] towards an emptier spot"* (P14). For 5 participants, the presence of real people in the room would also have affected the preferred placement: *"If there was a person there, I wouldn't want someone else - an avatar - on top of them"* (P7).

4 Discussion

Throughout our study, every F-Formation was mentioned as a favourite by at least one participant in one condition. This variety exemplifies the diverse and contrasting earlier literature [41, 49, 51] and the variety of factors found to influence group IPD and formations, emphasizing the need for AR collaboration tools to offer flexible avatar placement options. Previous research on social layouts in the real world has shown mixed results. While some studies suggested that negotiation tasks, being conversational in nature, favour C2C arrangements [15, 16, 49], others linked the competitive nature of negotiation to a preference for F2F layouts [11, 40, 41, 51]. Our results showed a preference for both F2F and sideF2F formations driven by ergonomics and collaborator visibility, as indicated by prior studies [1, 40, 47]. However, F2F elicited mixed reactions due to its perceived formality, adversarial undertones, and its tendency to highlight the limitations of headsets [5] and avatars [2]. In contrast, sideF2F received universally positive feedback, emerging as a highly favourable and previously under-explored option.

A second contribution of our study consisted in verifying the impact of the physical context on collaborator placement preferences. Previous research found that users prefer avoiding physicality conflicts with the environment [3, 30, 31, 44] and adhering to semantic coherence when placing virtual content in the physical context [12, 31, 34], such as placing a chat window positioned on a sofa, acting as a stand-in for an interlocutor seated beside them. These strands of research suggest that participants would consistently see avatar placements that overlapped with the box as unfavourable. However, our studies revealed that the physical context is nuanced. While most users did consider the box as an anti-affordance, over a quarter of them viewed it as a *valuable background over which to place their collaborator*. Participants that shared this view reported appreciating its visually simple, flat surface as helping them to focus, and that surface looking like the back-rest of a chair, helpful to ground the avatar, improving their immersion. At the same time, the chairs and the tree were more predictable, with the tree strongly considered an anti-affordance and the chairs clear affordances.

The contrasting views regarding the box highlight that there are more factors influencing preferred placement than previously considered [32, 47]. From the interviews three main factors emerged:

- **Semantic Coherence** - Most participants avoided the cognitive dissonance arising from virtual content interacting unnaturally with the physical context. However, our findings suggest that semantic coherence may be just as important but more *flexible* than previously thought [10, 12, 34]. Simple, evocative, or easily abstractable objects, such as cardboard boxes, can be reinterpreted to overcome concerns about

physicality conflicts. Semantically unambiguous objects, like a chair and a tree, cannot, evoking consistent interpretations as affordances or anti-affordances, respectively.

- **Visual Complexity** - While previous literature highlighted the importance of content legibility [10, 14] and of depth mismatches [3, 30, 31], our interviews highlighted the importance of visual simplicity of the physical context when assessing where to place virtual collaborators. Participants appreciated seeing their collaborator over empty surfaces, even if the physicality conflict was considered a drawback. Users mentioned feeling more focused and more immersed, important factors for collaborative platform designers to keep in mind.
- **Social Etiquette** - As suggested by previous research, users avoid placing virtual content between themselves and real people in the physical environment [4, 10, 39], reflecting an awareness of social etiquette. Our interviews revealed that this sense of etiquette extends to the placement of remote collaborators' avatars within the physical context. Participants often refrained from positioning avatars in certain locations (e.g., on the tree, between chairs) out of consideration for the remote collaborators, treating them *as if* they were physically present.

5 Conclusion and Future Work

We conducted two user studies to investigate local users' preferred social layouts when positioning a remote collaborator's avatar in Augmented Reality (AR) across physical settings with different objects. This is important for designing AR multi-user platforms and tools that will support collaboration in rich, messy, everyday environments.

This paper presents our first step in investigating the impact of the physical context on user preferences regarding collaborator avatar placement. Future research will further delineate which factors and objects of the physical context affect user preferences and quantitatively measure their impact. We will also investigate the role that task type has on social layout preference so as to expand our understanding towards other conversation-based tasks used in the workplace.

Our study found participants favoured frontal and sideF2F formations for ergonomic benefits and clear communication, though F2F elicited mixed reactions due to its formal and adversarial undertones. SideF2F emerged as a consistently positive option for avatar placement. The physical context influenced preferences, with chairs being seen as clear affordances, trees as anti-affordances, and the box eliciting mixed reactions. Three factors shaped these preferences: semantic coherence, visual complexity, and social etiquette. These findings underscore the need for AR tools that support flexible, context-sensitive avatar placement.

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A Appendices

A.1 Qualitative Code-books by Theme

A.1.1 Study 1.

Categories	Codes	Sub-Codes	#
Preferred F-Formations			225
F-Formations			114
F2F			64
sideF2F			26
C2C			15
SBS			9
Collaborator visibility			35
Visibility - Frontal is best			10
Visibility - F2F is best			11
Visibility - sideF2F is best			5
Visibility - crucial for negotiation			5
Visibility - F2F's great visibility breaks immersion			4
Visibility - Box helped			1
Formality/Intensity			26
Formality - F2F was too adversarial			11
Formality - F2F was too formal			8
Formality - sideF2F was just right			3
Formality - Frontal are best			2
Formality - Lateral too casual			2
Ergonomics			23
Ergonomics - F2F is best			8
Ergonomics - Frontal is best			7
Ergonomics - sideF2F is best			1
Ergonomics - C2C is best			1
Ergonomics - Lateral is bad			1
Ergonomics - Impact of headset			1
Personal space			23
Personal space - Close positions too uncomfortable			8
Dependent on social factors			7
Personal space - Far positions too uncomfortable			4
Personal space - sideF2F best			4
Focus			4
Focus - Frontal positions are best			2
Focus - C2C is best			1
Focus - sideF2F is best			1

Table 1: Table displaying the qualitative code-book theme *Preferred F-Formations*, with codes and sub-codes, gathered from the interview data of Study 1.

Categories	Codes	Sub-Codes	#
Physical object and Preferences			139
	Object was an anti-affordance		68
		Anti-affordance - Avoided the box	22
		Anti-affordance - Cognitive dissonance	22
		Anti-affordance - Visual dissonance	13
		Anti-affordance - Physicality conflict breaks immersion	6
		Anti-affordance - Social etiquette	5
		Anti-affordance - Physicality conflict looked weird	2
		Anti-affordance - Avoiding object helped focus	2
		Anti-affordance - Physicality conflict is unprofessional	2
	Object was an affordance		25
		Affordance - Box was a physical anchor	12
		Affordance - Box was like a chair	5
		Affordance - Box was like a background	5
		Affordance - Helps focus	3
		Affordance - Box was like a frame	2
	Plan B		14
		PlanB - Closest option to object	9
		PlanB - Symmetrical option to object	5
		PlanB - Mid-way between object and me	1
	Named other possible anti-affordances		14
	Object affordability depends on visual complexity		9
	No impact		9
		No impact - Did not mind physicality conflict	4
		No impact - Physicality conflict was bearable	4
		No impact - Depended on the proximity	1

Table 2: Table displaying the qualitative code-book theme *Physical object and Preferences*, with codes and sub-codes, gathered from the interview data of Study 1.

Categories	Codes	Sub-Codes	#
Collaborator characteristics			108
	Intimacy		47
		Intimacy matters	40
		Intimacy does not matter	7
	Social status		33
		Social status matters	24
		Social status does not matters	9
	Gender		28
		Gender does not matter	16
		Gender matters	12

Table 3: Table displaying the qualitative code-book theme *Collaborator characteristics*, with codes and sub-codes, gathered from the interview data of Study 1.

Categories	Codes	Sub-Codes	#
Task type			90
	Task type matters		35
	Competition		24
		Competition - Negotiation needs frontal formations	22
		Competition - Negotiation needs less frontal formations	1
		Competition - Depends on intimacy	1
	Cooperation		18
		Cooperation - Brainstorming needs closer formations	9
		Cooperation - When brainstorming position doesn't matter	6
		Cooperation - Brainstorming requires less eye contact	3
	Task type does not matter		8
	Mobility context made a difference		5

Table 4: Table displaying the qualitative code-book theme *Task type*, with codes and sub-codes, gathered from the interview data of Study 1.

A.1.2 Study 2.

Categories	Codes	Sub-Codes	#
Physical objects and Preferences	Object was an anti-affordance		210
			88
		Anti-affordance - Avoided the tree	36
		Anti-affordance - Cognitive dissonance	34
			10
		Anti-affordance - Social etiquette	8
		Anti-affordance - Avoiding object helped focus	7
		Anti-affordance - Physicality conflict breaks immersion	2
	Object was an affordance		37
		Affordance - Aimed for chairs/chairs were an anchor	29
		Affordance - Chairs helped immersion	5
		Affordance - Helps focus	2
		Affordance - Tree was a physical anchor	1
	Best position is the most realistic position		32
		Comparison between box and tree	17
		Tree more impactful	13
		No difference	3
	Plan B	Box more impactful	1
			14
		PlanB - Closest option to object	11
		PlanB - Symmetrical option to object	2
	Object affordability depends on visual complexity	PlanB - Mid-way between object and me	1
			12
	No impact		10
		No impact - Could bear with avatar not on chair	7
		No impact - Did not aim for chairs	3

Table 5: Table displaying the qualitative code-book theme *Physical objects and Preferences*, with codes and sub-codes, gathered from the interview data of Study 2.

Categories	Codes	Sub-Codes	#
Preferred F-Formations	Collaborator visibility		69
		Visibility - Frontal is best	22
		Visibility - F2F is best	14
		Visibility - sideF2F is best	5
		Visibility - crucial for negotiation	2
Ergonomics			1
		Ergonomics	21
		Ergonomics - sideF2F is best	6
			5
		Ergonomics - F2F is best	4
		Ergonomics - Lateral is bad	4
		Ergonomics - C2C is best	1
Formality/Intensity		Ergonomics - Impact of headset	1
			9
		Formality - F2F was too adversarial	3
		Formality - F2F was too formal	3
		Formality - sideF2F was just right	2
			1
		Personal space	6
Personal space		Personal space - Far positions too uncomfortable	3
		Personal space - sideF2F best	2
		Personal space - C2C best	1
			1
			1
Task type matters		Task type matters	11
		Negotiation needs frontal formations	7
		Brainstorming needs closer formations	4

Table 6: Table displaying the qualitative code-book theme *Preferred F-Formations*, with codes and sub-codes, gathered from the interview data of Study 2.

Categories	Codes	Sub-Codes	#
Physical background and Preferences	Background mattered		45
		Background - Window	36
		Background - Size/Shape of room	16
	Other elements that would have mattered		7
		Other - Presence of other people matters	9
		Other - Presence of objects on table matters	5
			4

Table 7: Table displaying the qualitative code-book theme *Physical background and Preferences*, with codes and sub-codes, gathered from the interview data of Study 2.